

Physics You Can Feel: A Pseudo-Haptic VR Sandbox for Experiencing Mechanics and Material Properties

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Abstract

Physical intuition about mass, volume, density, and friction, and about simple machines such as levers and pulleys, is hard to build from equations and 2D diagrams, and the demonstrations that would make it concrete are often impractical or unsafe in a classroom. We propose an immersive virtual-reality sandbox in which learners create, hold, weigh, and manipulate objects and simple machines. Because commodity headsets cannot render true weight, we use pseudo-haptic cues, modulating control-to-display gain and visual lag, so that a heavier object feels heavier and a more heavily loaded lever feels harder to move; the cue tracks mass and load, while density is revealed through the volume of equal-mass blocks and through whether they float or sink. The application is pure VR with no in-scene agent; agentic AI is instead the development workflow, an AI coding agent that helps implement the Unity project. We evaluate the system against closed-form physics and run a small within-subjects study testing whether the weight cue improves how accurately learners predict outcomes that depend on mass, density, and load.

Keywords

3D user interfaces; virtual reality; embodied learning; pseudo-haptics; physics education; agentic AI; AI-assisted development

1 Introduction and Motivation

Introductory physics rests on relationships that students can state long before they can predict with them. Density relates mass to volume, buoyancy determines whether an object floats, friction resists sliding in proportion to the normal load, and simple machines such as levers and pulleys trade force for distance. The usual failure is not arithmetic but intuition. Asked whether a kilogram of gold or a kilogram of ice is larger, which of two blocks slides farther down a ramp, or where a load balances on a lever, many learners fall back on misconceptions such as “heavier means bigger.” Textbooks and 2D simulations show the formula and the result, yet they do not let a student *feel* the difference.

Doing the demonstration in real life is harder than it sounds: the comparisons that isolate a single variable are exactly the ones that are awkward, expensive, or unsafe to stage, since matched-mass materials are not classroom stock and friction is hard to hold constant. Virtual reality can

present these controlled manipulations on demand, but a handheld controller weighs the same whatever it represents, so the sensation many lessons depend on is missing. The open question is whether a spatial interface can make physical properties and mechanical behavior *felt* well enough to matter, given that the dominant channel for weight and force is the one current hardware cannot deliver.

2 Related Work

Our sandbox sits at the intersection of four lines of research: immersive virtual reality (VR) for science learning, the difficulty of rendering weight and force on commodity hardware, the pseudo-haptic techniques that stand in for that missing sensation, and the interaction-design foundations that make direct manipulation legible. We review each in turn and close by stating what distinguishes our approach.

2.1 Immersive VR for Science Learning

The pedagogical case for immersive VR rests less on visual fidelity than on the sense of being present. Slater and Sánchez-Vives frame this through the place illusion (the feeling of being in the virtual space) and the plausibility illusion (the feeling that events are really happening), which together lead users to respond to virtual events as if they were real [8]. For a physics sandbox this matters directly. If a learner treats a virtual block and lever as real objects, then manipulating them can carry the intuitive weight that a 2D simulation cannot.

Empirical work in physics education adds a consistent caveat: immersion helps, but only when the learning experience is designed for it. A recent meta-analysis by Wu et al. across many head-mounted display studies finds that immersive VR yields a positive but moderate overall effect on learning performance, and that this effect is strongly moderated by factors such as subject domain, learner control, and how the experience is designed [12]. Tsivitanidou, Georgiou, and Ioannou report on an inquiry-based immersive VR physics activity from two angles: the learning experience they observe conceptual gains alongside an interaction with learners’ attitudinal profiles [9], and the design of that same experience, which they document as a worked case of learning-experience design [3]. We read this pair as evidence that immersive VR physics learning is feasible and can produce conceptual gains; it establishes that a VR physics activity can work, though it

does not itself isolate felt weight or the effect of a haptic cue. One common way to scaffold such experiences is an embedded pedagogical agent. Makransky et al. studied one such agent and found a gender-matching effect: a virtual instructor aided learning most when its apparent gender matched the learner, a specific interaction rather than a general endorsement of pedagogical agents [6]. Taken together, these studies establish that immersive physics learning is promising but design-sensitive. The value comes from what the learner is asked to do and how the task is scaffolded, not from the headset itself. We take a deliberately different scaffolding stance. Rather than an in-scene agent, we scaffold through scene and task design, using tightly scoped scenarios that isolate a single variable so that the cause of each outcome is unambiguous.

2.2 The Missing Channel: Rendering Weight and Force

The central obstacle for a physics-of-materials experience is that the very quantity the lessons depend on, namely weight and the force needed to move a loaded mechanism, is the one commodity that VR cannot deliver. A handheld controller weighs the same whatever it represents. Wang et al. survey the state of haptic display for VR and make clear that genuine, general-purpose force feedback remains far from consumer reach, constrained by actuation, workspace, and grounding requirements that headset-and-controller setups do not meet [11]. Their survey frames the design problem for us precisely. If we want a heavier object to feel heavier on a Quest 2, we cannot wait for the hardware to render the force, and must instead substitute perception for physics.

2.3 Pseudo-Haptics: Substituting Perception for Physics

Pseudo-haptics addresses this gap by exploiting the dominance of vision over proprioception. Manipulating the visual consequences of a motion, such as the control-to-display gain or a small lag between hand and object, induces a felt sense of weight, stiffness, or resistance without any force actuator. The foundational demonstration comes from Lécuyer et al., who showed that altering the visual response of an isometric input device makes users perceive force and stiffness that no actuator produces, establishing that a control-to-display manipulation alone can stand in for force feedback [5]. Ujitoko and Ban survey the design space and applications of pseudo-haptics, cataloguing how such visual manipulations reliably produce haptic sensations and how they have been applied [10]. Their taxonomy is the methodological backbone for how we map mass and mechanical load onto gain and visual lag.

Two results ground our specific cue, the felt *weight* of a held object. Dominjon et al. manipulated the control/display ratio of a manipulated object in a virtual environment and showed that this ratio directly changes its perceived mass, so that the same physical motion can be made to feel heavier or lighter by scaling its visual displacement [2]. Rietzler et al. carried

the same idea to consumer VR: by introducing perceivable offsets between the tracked hand and the rendered object, they elicited a reliable sensation of weight on a commodity headset with no force hardware [7]. Together these establish that a gain or tracking-offset manipulation can render felt weight on exactly the class of device we target, and they fix the design variable for us: the cue must track an object’s *mass*, the quantity that determines weight, rather than its density. These techniques are not limited to lifting a single object. The same principle of modulating the visual outcome of an effort extends naturally to the resistance of operating a machine, such as a lever that grows harder to move as its load increases.

2.4 Foundations of 3D Interaction

Whatever the illusion, the learner must first be able to reach out, grab, and place objects fluidly, or the physics never gets probed. We ground our interaction design in Bowman et al.’s canonical treatment of 3D user interfaces, which formalizes selection, manipulation, and travel as the core tasks of spatial interaction and catalogues the techniques for each [1]. This foundation informs the grab-and-place model at the heart of the sandbox and keeps our contribution, the pseudo-haptic property mapping, layered on top of well-understood manipulation techniques rather than reinventing them.

2.5 What Is Different About Our Approach

Two gaps emerge from this literature. First, immersive physics studies consistently deliver visual and interactive experiences but rarely give learners a genuine felt contrast in weight between objects. A student may see that two blocks differ, but does not feel one resist more than the other. Second, pseudo-haptic studies convincingly render the felt weight of an isolated object [2, 7], but seldom span both material properties (density, buoyancy, friction) and simple machines (levers, pulleys) within one coherent environment and interaction model. Our sandbox targets this intersection. We use pseudo-haptic gain and visual-lag modulation as a unifying mechanism that makes an unrenderable physical quantity perceptible across a family of controlled, single-variable scenarios, and we extend the same mechanism from holding a heavier object to feeling a loaded lever push back. Crucially, the cue tracks *mass* for held objects and *load* for the lever, the quantities that actually determine weight and effort, so that density is taught where it belongs, through the volume difference between equal-mass blocks and through float-or-sink behavior, rather than being conflated with heaviness. Because our contribution is an interaction technique for making a physical quantity felt, we evaluate it at two levels: a system level, checking that the cues behave monotonically with the underlying physics and that each scenario matches its closed-form prediction; and a human level, a within-subjects study testing whether the weight cue changes how accurately

learners predict outcomes that depend on mass, density, and load.

3 Approach

We will build a VR “physics sandbox” in which the learner constructs and probes scenarios and the interface makes the governing quantity perceptible. Early scenarios target material properties: pick up two blocks and feel which is heavier on a balance, drop them in water to see buoyancy, and drag them across a ramp to compare friction. Density is made visible rather than felt: two *equal-mass* blocks of different materials differ in size, so the denser one is visibly smaller, and the same pair floats or sinks by density in the tank. The same interaction model extends to mechanical devices, such as placing weights on a lever to find its balance point or routing a rope through a pulley to feel the reduced effort. In each case a single variable is isolated so the outcome has a clear cause. Weight and resistance are conveyed through pseudo-haptics, modulating control-to-display gain and visual lag so that a heavier block feels heavier and a more heavily loaded lever feels harder to move [2, 7, 10]. The cue tracks an object’s *mass*, the quantity that determines its weight, rather than its density: equal-mass blocks weigh the same whatever their material, so making a denser block feel heavier would teach the very “heavier means bigger” misconception the sandbox sets out to correct. This is where the project complies with HCI research in **Virtual Reality**: the contribution is an interaction technique for making an unrenderable physical quantity perceptible through direct, embodied manipulation.

The second aspect is the **agentic-AI** development workflow, and we are deliberate about the distinction the assignment draws. The finished application has no in-scene chatbot for the learner to talk to. Instead, agentic AI is how the software is built: an AI coding agent works in the project with real tools, editing Unity C# scripts, running builds, reading compiler and test output, and iterating toward a working scene over many steps rather than emitting one block of code from a single prompt. It plans, uses tools, observes results, and self-corrects toward a goal. We report what it implemented, where it needed human correction, and how the workflow shaped the build. The professor has confirmed that a pure VR project built this way satisfies the assignment.

4 System Vision and Technology

When it works, the demo is a first-person VR scene with a ground plane, a work area, and a menu for spawning materials and mechanical devices (Figures 1 and 2). The learner grabs objects with hand-tracking or controllers and feels a heavier one resist motion or a loaded lever push back; a side panel shows each object’s properties and the measured outcome of an interaction. Early scenarios cover density, buoyancy, and friction, with levers, pulleys, and projectiles added on the same framework as the project grows.

We will build in **Unity 6** with the **XR Interaction Toolkit** and **OpenXR**, in **C#**, targeting the **Meta Quest**

2 (hand-tracking and controllers). Physics uses Unity’s rigid-body engine with per-material density and friction parameters and standard joints for levers and pulleys. Development uses an agentic-AI coding workflow, an AI agent that edits scripts, runs builds, and reads compiler output while iterating toward each scene. We already have a Quest 2 and a Unity development machine.

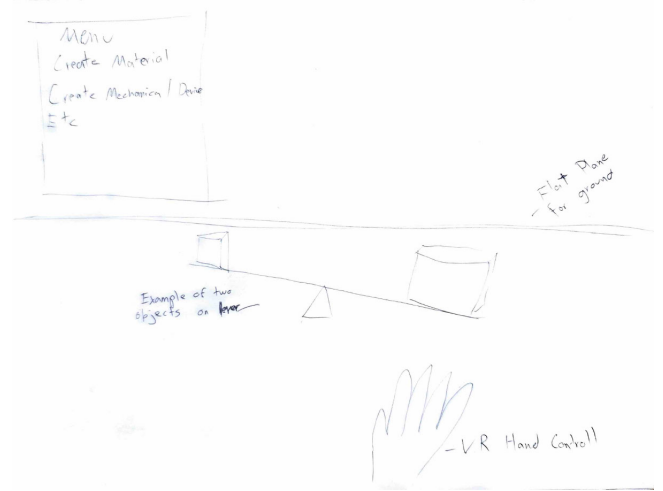


Figure 1: Hand-drawn concept sketch. From the learner’s first-person view: a menu panel (create a material, create a mechanical device, and so on), two objects balanced on a lever over a fulcrum on a flat ground plane, and a tracked VR hand for direct grab-and-place interaction.

5 Prototype Validation

Before any human data is collected, the prototype must be shown to behave correctly, and these system-level checks are a precondition for a meaningful study. **Physics fidelity:** automated checks compare each scenario against its closed-form prediction within a 5% tolerance, for example that equal-mass blocks show the correct relative volumes, objects float or sink by density, ramp sliding follows the friction coefficient, and a lever balances at the computed load position. **Pseudo-haptic behavior:** the cues must be monotonic, a heavier object always feeling heavier and a more heavily loaded lever always harder to move, with gain and lag mapping to mass and load in the intended direction across the full range of materials and loads. **Performance:** the scene must hold 72 Hz on the Quest 2 so the cue is not confounded by dropped frames. **Development workflow:** because agentic AI is our build method, we report which parts of the implementation the coding agent produced through tool use and iteration and where human correction was needed, drawn from a task log kept from the first day of development. Passing these checks establishes that the two builds differ only in the presence of the weight cue, which is what the study in Section 6 requires.



Figure 2: AI-enhanced render of the same scene (generated with Google Gemini from the sketch in Figure 1). A labeled steel block and concrete cylinder rest on a lever, showing how the learner reads object properties and manipulates them by hand while the floating menu spawns new materials and mechanisms.

6 Methodology

Making a physical quantity *felt* is only worthwhile if it changes what learners understand, so beyond validating the system we conducted a controlled study with human participants. It asked one question: does a pseudo-haptic weight cue, added to direct manipulation in VR, change how accurately learners predict outcomes that depend on mass, density, and load, compared with the same manipulation without the cue? We report the study using the standard structure of participants, apparatus, procedure, and design.

6.1 Participants

Twelve participants (the class minimum for a repeated-measures design) were recruited from non-physics courses, targeting students who had not taken university physics so that measurable misconceptions were present to begin with. Recruitment used the standard class data-collection form. The project was not intended for outside publication, so no IRB submission was required; were publication pursued later, an IRB with a faculty principal investigator would be arranged before any recruitment began. Participation took about 40 minutes, and participants could stop at any time.

6.2 Apparatus

The study used a Meta Quest 2 running the sandbox described above, built in Unity 6 with the XR Interaction Toolkit and OpenXR. Two builds were prepared that were visually identical and differed only in the weight cue. In the cue-on build, control-to-display gain and visual lag tracked an object’s mass and the lever’s load, so that a heavier object felt heavier and a more heavily loaded lever felt harder to

move; the cue-off build used plain grab-and-place with no such modulation. Held objects were made kinematic while grasped, the cue was applied to the displayed position, and each object was handed back to the physics engine on release, so that the simulation and the cue did not fight. Before any session, the prototype passed the validation checks of Section 5: each scenario matched its closed-form prediction within 5%, the cue was monotonic across the full range of materials and loads, and the scene held 72 Hz. The pre/post test, heaviness judgments, and questionnaire were administered on a laptop between VR blocks.

6.3 Procedure

Each participant completed a pre-test, then a short training scene to learn grab-and-place interaction, then two blocks of four scenes each (blocks covered same-mass block comparison, buoyancy in a tank, sliding on a ramp, and balancing a lever). Each block was followed by a post-test, and the session ended with a brief interview about what felt different between the two blocks. After each pair of blocks, participants also made a two-alternative forced-choice heaviness judgment (which block felt heavier) and completed a five-item presence-and-comfort questionnaire.

6.4 Design

The experiment was within subjects with one independent variable, the *weight cue* (on vs. off). Cue order was counterbalanced across participants, and two matched scenario sets (A and B) were paired with the two cue conditions so that no participant saw the same scene twice; set was counterbalanced against cue condition.

The primary dependent variable was *prediction accuracy* on a short misconception test given before and after each block. Its six items followed the style of the Force Concept Inventory [4]: which of two equal-mass blocks is larger, which floats, which slides farther down a ramp, where a lever balances, and two transfer items using materials not present in the sandbox. Secondary dependent variables were the forced-choice heaviness judgment (a manipulation check that the cue was perceived at all), task completion time, and the presence-and-comfort score.

We tested three hypotheses:

- **H1:** the gain in prediction accuracy from pre-test to post-test was larger with the cue on than with the cue off.
- **H2:** heaviness judgments were above chance with the cue on and at chance with the cue off (the manipulation check).
- **H3:** reported presence was higher with the cue on.

H1 was tested with a 2×2 repeated-measures ANOVA on accuracy (cue $\times$ pre/post), with the interaction term as the effect of interest; H2 with a binomial test against chance in each condition; and H3 with a paired *t*-test on the presence score.

7 Deliverables

By **mid-November**:

- A Unity 6 scene on Quest 2 with grab-and-place interaction for at least four materials, a working balance, a water tank, and a ramp, plus one mechanical device (a lever), all with pseudo-haptic weight and force modulation (gain and visual lag).
- Automated physics-fidelity checks comparing the simulation against closed-form predictions for density, buoyancy, friction, and the lever.
- In-scene logging of interactions and outcomes to file.

Toward the **early-December** presentation:

- An expanded sandbox adding at least one more concept (for example a pulley or projectile launcher) built on the same interaction and physics framework, delivered as two builds (cue on and cue off) for the study.
- The within-subjects study of Section 6 run with twelve non-physics students, with the pre/post misconception test, heaviness manipulation check, and presence questionnaire analyzed as described.
- A recorded demonstration video walking through each concept on-device.
- The paper extending this proposal, documenting the design, the system and human evaluation, and a report on the agentic-AI development workflow used to build it, grounded in the day-one development log.

Roles: This is a solo project (group CS567_SVEN). Sven Skoglund is responsible for all work, including the Unity interaction and pseudo-haptics, the physics-fidelity checks, the demonstration, and directing the agentic-AI development workflow.

8 Conclusion

We are building a VR physics sandbox that turns abstract relationships, from density and friction to the mechanics of levers and pulleys, into things a learner can feel and test through direct, embodied manipulation. The application is pure VR with no in-scene agent; agentic AI is the workflow used to build it. The class should expect a working Quest 2 demonstration, automated checks that the simulation matches the underlying physics, a within-subjects study of whether the weight cue improves learners' physical predictions, and a report on how the agentic-AI development process shaped the result.

Use of AI

Generative AI (a large language model) was used to help locate candidate peer-reviewed references and to critique the argument and grammar of drafts. Google Gemini generated the enhanced mockup in Figure 2 from the author's hand-drawn sketch (Figure 1). The implementation is planned as an agentic-AI development effort in which an AI coding agent assists in writing and iterating on the Unity project under the author's direction, and this will be documented in the final

paper. All prose in this proposal was reviewed and revised by the author, and every cited work was read and verified by the author.

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